

WORK №2. Optimal control design with free and fixed initial state point

The order of the work

1. Design the optimal controller according to the given cost function assuming that $q=1, r=1$ **and the end** $x_1(t_f), x_2(t_f)$ **is FREE (do not use** $x_1(t_f), x_2(t_f)$ **)** and simulate its work within the given time interval.

2. Design the optimal controller according to the given cost function assuming that $q=0, r=1$ **and the end** $x_1(t_f), x_2(t_f)$ **is FIXED and given in the table** and simulate its work within the given time interval.

3. Plot the control, state, and the cost function evolution for cases 1 and 2.

4. Plot the cost function for slightly changed and unchanged controller parameters at one picture for cases 1 and 2. Make conclusions.

№	Plant	Cost function	The boundary conditions
1	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -4x_1 - 5x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 5)^2 + (x_2(t) - 1)^2 \right] \Big _{t_f=1} + 4r \int_0^1 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(1) = 5, x_2(1) = 1$
2	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -10x_1 - x_2 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 7)^2 \Big _{t_f=2} + \frac{r}{2} \int_0^2 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(2) = 7, x_2(2) = 0$
3	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -x_1 - 9x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 8)^2 + (x_2(t) - 3)^2 \right] \Big _{t_f=4} + \frac{r}{2} \int_0^4 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(4) = 8, x_2(4) = 3$
4	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -3x_1 - 6x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 1)^2 + (x_2(t) - 9)^2 \right] \Big _{t_f=6} + \frac{3r}{2} \int_0^6 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(6) = 1, x_2(6) = 9$

5	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -8x_1 - 7x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 3)^2 + x_2^2(t) \right] \Big _{t_f=5} + \frac{5r}{2} \int_0^5 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(5) = 3, x_2(5) = 0$
6	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -3x_1 - 6x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 9)^2 + (x_2(t) - 1)^2 \right] \Big _{t_f=9} + \frac{3r}{2} \int_0^9 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(9) = 9, x_2(9) = 1$
7	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -x_1 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 1)^2 + x_2^2(t) \right] \Big _{t_f=2} + \frac{r}{2} \int_0^2 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(2) = 1, \\ x_2(2) = 0$
8	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -3x_1 - 6x_2 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 7)^2 \Big _{t_f=\pi} + \frac{r}{2} \int_0^\pi u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(\pi) = 7, x_2(\pi) = 0$
9	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -7x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 3)^2 \Big _{t_f=1} + \frac{r}{2} \int_0^1 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(1) = 3, x_2(1) = 0$
10	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -2x_1 - 9x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 2)^2 + x_2^2(t) \right] \Big _{t_f=6} + r \int_0^6 3u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(6) = 2, x_2(6) = 0$
11	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -5x_2 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 1)^2 \Big _{t_f=9} + \frac{5r}{2} \int_0^9 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(9) = 1, x_2(9) = 0$
12	$\begin{cases} \dot{x}_1 = 4x_2, \\ \dot{x}_2 = x_1 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 1)^2 + x_2^2(t) \right] \Big _{t_f=9} + \frac{7r}{2} \int_0^9 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(9) = 1, \\ x_2(9) = 0$
13	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -3x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 2)^2 \Big _{t_f=\pi} + \frac{r}{2} \int_0^\pi u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(\pi) = 2, \\ x_2(\pi) = 0$

14	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 1)^2 \Big _{t_f=1} + \frac{r}{2} \int_0^1 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(1) &= 1, x_2(1) = 0 \end{aligned}$
15	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 1)^2 \Big _{t_f=\pi} + \frac{r}{2} \int_0^\pi u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(\pi) &= 1, x_2(\pi) = 0 \end{aligned}$
16	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -4x_1 + u \end{cases}$	$\begin{aligned} J &= \frac{q}{2} \left[(x_1(t) - 10)^2 + x_2^2(t) \right] \Big _{t_f=1} \\ &+ \frac{r}{2} \int_0^1 u^2(\tau) d\tau \end{aligned}$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(1) &= 10, x_2(1) = 0 \end{aligned}$
17	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 1)^2 \Big _{t_f=5} + \frac{r}{2} \int_0^5 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(5) &= 1, x_2(5) = 0 \end{aligned}$
18	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -6x_1 - 5x_2 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 1)^2 \Big _{t_f=4} + \frac{r}{2} \int_0^4 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(4) &= 1, x_2(4) = 0 \end{aligned}$
19	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -9x_1 + u \end{cases}$	$\begin{aligned} J &= \frac{q}{2} \left[(x_1(t) - 2)^2 + x_2^2(t) \right] \Big _{t_f=\pi/2} \\ &+ \frac{r}{2} \int_0^{\pi/2} u^2(\tau) d\tau \end{aligned}$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(\pi) &= 2, \\ x_2(\pi/2) &= 0 \end{aligned}$
20	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -x_1 - 2x_2 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 100)^2 \Big _{t_f=1} + \frac{r}{2} \int_0^1 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(1) &= 100, \\ x_2(1) &= 0 \end{aligned}$
21	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = x_1 + x_2 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 1)^2 \Big _{t_f=3} + \frac{r}{2} \int_0^3 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(3) &= 1, \\ x_2(3) &= 0 \end{aligned}$
22	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -8x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 5)^2 \Big _{t_f=3} + \frac{r}{2} \int_0^3 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(3) &= 5, x_2(3) = 0 \end{aligned}$
23	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -6x_1 - 3x_2 + u \end{cases}$	$\begin{aligned} J &= \frac{q}{2} \left[(x_1(t) - 6)^2 + (x_2(t) - 2)^2 \right] \Big _{t_f=6} \\ &+ \frac{7r}{2} \int_0^6 u^2(\tau) d\tau \end{aligned}$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(6) &= 6, x_2(6) = 2 \end{aligned}$

24	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -7x_2 + u \end{cases}$	$J = \frac{q}{2}(x_1(t) - 2)^2 \Big _{t_f=8} + \frac{r}{2} \int_0^8 4u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(8) = 2, x_2(8) = 0$
25	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -x_1 + u \end{cases}$	$J = \frac{q}{2}(x_1(t) - 2)^2 \Big _{t_f=1} + \frac{r}{2} \int_0^1 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(1) = 2, x_2(1) = 0$
26	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -4x_1 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 3)^2 + x_2^2(t) \right] \Big _{t_f=\pi/6} + \frac{r}{2} \int_0^{\pi/6} u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(\pi/6) = 3, \\ x_2(\pi/6) = 0$
27	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -7x_1 - 5x_2 + u \end{cases}$	$J = \frac{q}{2}(x_1(t) - 9)^2 \Big _{t_f=4} + \frac{r}{2} \int_0^4 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(4) = 9, \\ x_2(4) = 0$
28	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = 6x_1 + 4x_2 + u \end{cases}$	$J = \frac{q}{2}(x_1(t) - 5)^2 \Big _{t_f=2} + r \int_0^2 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(2) = 5, \\ x_2(2) = 0$
29	$\begin{cases} \dot{x}_1 = 4x_2, \\ \dot{x}_2 = -3x_1 - 7x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 7)^2 + x_2^2(t) \right] \Big _{t_f=6} + \frac{3r}{2} \int_0^6 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(6) = 7, \\ x_2(6) = 0$
30	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -5x_1 + u \end{cases}$	$J = \frac{q}{2}(x_1(t) - 4)^2 \Big _{t_f=5} + \frac{r}{2} \int_0^5 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(5) = 4, \\ x_2(5) = 0$
31	$\begin{cases} \dot{x}_1 = 4x_2, \\ \dot{x}_2 = u \end{cases}$	$J = \frac{q}{2}(x_1(t) - 3)^2 \Big _{t_f=5} + \frac{5r}{2} \int_0^5 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(5) = 3, x_2(5) = 0$
32	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -3x_1 - 6x_2 + u \end{cases}$	$J = \frac{q}{2}(x_1(t) - 10)^2 \Big _{t_f=1} + \frac{3r}{2} \int_0^1 u^2(\tau) d\tau$	$x_1(0) = x_2(0) = 0, \\ x_1(1) = 10, \\ x_2(1) = 0$

33	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = 3x_1 + 2x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 4)^2 + x_2^2(t) \right] \Big _{t_f=8} + \frac{9r}{2} \int_0^8 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(8) &= 4, \\ x_2(8) &= 0 \end{aligned}$
34	$\begin{cases} \dot{x}_1 = 2x_2, \\ \dot{x}_2 = 8x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 5)^2 \Big _{t_f=4} + \frac{3r}{2} \int_0^4 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(4) &= 5, \\ x_2(4) &= 0 \end{aligned}$
35	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -6x_1 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 3)^2 + x_2^2(t) \right] \Big _{t_f=9} + r \int_0^9 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(9) &= 3, \\ x_2(9) &= 0 \end{aligned}$
36	$\begin{cases} \dot{x}_1 = 11x_2, \\ \dot{x}_2 = u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 3)^2 \Big _{t_f=3} + \frac{r}{4} \int_0^3 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(3) &= 3, x_2(3) = 0 \end{aligned}$
37	$\begin{cases} \dot{x}_1 = 5x_2, \\ \dot{x}_2 = -6x_1 + 2u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 4)^2 + (x_2(t) - 5)^2 \right] \Big _{t_f=4} + \frac{r}{2} \int_0^4 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(4) &= 4, \\ x_2(4) &= 5 \end{aligned}$
38	$\begin{cases} \dot{x}_1 = 4x_2, \\ \dot{x}_2 = -11x_1 + 5u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 3)^2 + (x_2(t) - 6)^2 \right] \Big _{t_f=5} + \frac{r}{2} \int_0^5 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(5) &= 3, x_2(5) = 6 \end{aligned}$
39	$\begin{cases} \dot{x}_1 = 4x_2, \\ \dot{x}_2 = -7x_1 - x_2 + 6u \end{cases}$	$J = q \left[(x_1(t) - 2)^2 + (x_2(t) - 3)^2 \right] \Big _{t_f=\pi/2} + r \int_0^{\pi/2} u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(\pi/2) &= 2, \\ x_2(\pi/2) &= 3 \end{aligned}$
40	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -9x_1 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 4)^2 + x_2^2(t) \right] \Big _{t_f=\pi} + \frac{r}{2} \int_0^{\pi} u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(\pi) &= 4, x_2(\pi) = 0 \end{aligned}$

41	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -6x_1 + u \end{cases}$	$J = q(x_1(t) - 70)^2 \Big _{t_f=4} + r \int_0^4 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(4) &= 70, x_2(4) = 0 \end{aligned}$
42	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -5x_1 - 8x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 7)^2 + x_2^2(t) \right] \Big _{t_f=5} + r \int_0^5 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(5) &= 7, x_2(5) = 0 \end{aligned}$
43	$\begin{cases} \dot{x}_1 = 3x_2, \\ \dot{x}_2 = u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 4)^2 \Big _{t_f=2} + \frac{5r}{2} \int_0^2 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(2) &= 4, x_2(2) = 0 \end{aligned}$
44	$\begin{cases} \dot{x}_1 = 4x_2, \\ \dot{x}_2 = u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 9)^2 \Big _{t_f=3} + 4r \int_0^3 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(3) &= 9, x_2(3) = 0 \end{aligned}$
45	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -5x_1 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 6)^2 + (x_2(t) - 3)^2 \right] \Big _{t_f=5} + \frac{r}{2} \int_0^5 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(5) &= 6, \\ x_2(5) &= 3 \end{aligned}$
46	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -7x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 7)^2 \Big _{t_f=\pi/4} + \frac{r}{2} \int_0^{\pi/4} u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(\pi/4) &= 7, \\ x_2(\pi/4) &= 0 \end{aligned}$
47	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -3x_1 - 7x_2 + u \end{cases}$	$J = \frac{q}{2} \left[(x_1(t) - 13)^2 + x_2^2(t) \right] \Big _{t_f=6} + \frac{r}{2} \int_0^6 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(6) &= 13, x_2(6) = 0 \end{aligned}$
48	$\begin{cases} \dot{x}_1 = 3x_2, \\ \dot{x}_2 = 2x_1 + 4u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 4)^2 \Big _{t_f=9} + \frac{7r}{2} \int_0^9 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(9) &= 4, x_2(9) = 0 \end{aligned}$
49	$\begin{cases} \dot{x}_1 = 3x_2, \\ \dot{x}_2 = -x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 2)^2 \Big _{t_f=1} + \frac{r}{2} \int_0^1 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(1) &= 2, x_2(1) = 0 \end{aligned}$
50	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -x_1 + u \end{cases}$	$J = \frac{q}{2} (x_1(t) - 1)^2 \Big _{t_f=2} + \frac{r}{2} \int_0^2 u^2(\tau) d\tau$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(2) &= 1, \\ x_2(2) &= 0 \end{aligned}$